Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Knowledge-based AI assistance for automating regulatory compliance			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Knowledge services*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solutions*	*Management strategy / finance / trade services*	*HR Management / legal / accounting services*
Concept of Proposed Technology	Adhering to laws and regulations is a global challenge, both for companies and citizens. Regulations across the world are becoming more complex to tackle digitalization, AI and climate change ambitions. Laws and regulations can be hard to understand, not only for humans but also for systems, which makes it complex and labor intensive to check compliance. SMEs have a hard time adopting and continuously proving compliance to the laws and regulations, as they usually do not have the resources and/or expertise in-house. When SMEs (want to) do business internationally, the challenges of compliance checking increase even more, as in the EU for			

	example more than 1000 regulations are approved per year. A trustworthy manner to 1) formalize a law or set of rules and to 2) apply this formalization in context is required to automate compliance checking and to reduce the amount of work involved. To ease and automate the regulatory compliance checking processes that SMEs face, we propose a technology to make regulatory texts better understandable and accessible for each organization that needs to comply to laws, rules and/or policies. The AI-based technology consists of two modules: 1) semi-automatic creation of a formalization of laws and regulations, following the Rules as Code principles; 2) an AI assistant for question answering about the laws and regulations, using both state-of-the-art AI techniques and based on the created formalization and other knowledge-based AI-methods. The development of this technology requires a multi-disciplinary approach and combined knowledge on the law and various types of AI, as we specifically aim at integrating symbolic and sub-symbolic AI methods. This approach ensures that technology not only provides easy-to-read answers related to laws and regulations, but that the answers are also more factual and traceable to the statements in the laws or regulations. From TNO, we can bring the multi-disciplinary approach and combined knowledge, as we have extensive knowledge on Natural Language Processing to automatically create a formalization (references 1-4), on question-answering on legal questions (5), the proof-of-concept AI Assistant using the formalization (6), and legal formalizations (7-9). We can use our open source code base (10) as a head start. Our knowledge and code is already used by Dutch SMEs. The technology can be used by SMEs to check compliancy to the law or other regulations, but also by citizens that want to check the law. This technology is not bound to only the state law, but can for example also be used to check compliance against other legal documents, policies and domain or company specific regulations.
Rationale for Joint R&D Project	The trustworthy manner and technology to formalize a law or set of rules and to apply this formalization in context is currently lacking. AI-based solutions exist, but current (generative) AI-systems are heavily based on the English language and do not always work well when interacting in specific languages such Korean and Dutch. For automated regulatory compliance there is a need for trustworthy AI-systems that are factually correct (as far as possible) and able to provide explanations and reasoning, based on the specific language of the laws or regulations. In the Netherlands we have already created a formalization of laws and a proof-of-concept for the AI assistant for the Dutch language, which are currently tested and adopted in the Dutch context, specifically at the government level. With this R&D project we want to test and further develop the AI-

	modules for formalization and question answering of laws and regulations in other specific languages and thereby improving its reliability, by making the modules applicable to laws and regulations in the Korean language.
Key Words	1. Regulatory Compliance 2. Retrieval Augmented Generation (RAG) 3. Knowledge Graphs 4. Generative AI 5. AI Agents 6. Symbolic AI

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (TRL 4)	**End of the Project**	Level (TRL 7)
R&D Objectives	The overall project objective is to ease regulatory compliance checking for SMEs, by providing them with AI-based technology for making laws and regulations machine readable and interpretable, and with that facilitate low-threshold interaction with these laws. SMEs can use the technology to answer questions about the legislation and automate the assessment of their compliancy with the legislation. This leads to reduced compliance time and costs and with that higher operational efficiency. Examples of SMEs that related to these objectives are: - Law firms automating their support to clients - Manufacturing / logistics SME aiming to be compliant with legislation - Providers of digital health solutions - Technology or software provider (e.g. ERP system) which should be compliant with legislation. From the technology perspective, this asks for the improvement of the two modules of our AI-based solution: 1) semi-automatic module to create a formalization of the law, following the rules as code principles; 2) an AI assistant to answer questions about the law. The current modules are at TRL level 4 and only applicable to the Dutch law and language. In the project, we will adapt these modules to other specific laws and languages, specifically Korean legal texts and language, and their application. We also aim to perform an experiment with potential users of the system to reach TRL level 7.			
R&D Target Specifications	The AI assistant performance will be measures using several qualitative and quantitative metrics; we will use metrics such as precision, recall and accuracy in retrieving the right information from the legal sources (based on a ground truth dataset), as well as the correctness, completeness and fluency of the answers. Besides these metrics, users will evaluate the assistant on these metrics and on subjective metrics such as the SUS score (usability) and perceived performance. The formalization of the law will be tested using expert knowledge as well as it's application in the AI assistant: using the formalization will improve the performance of the AI assistant compared to only using the legal sources without the formalization – performance of both will be compared.			

(max 10 pages)

R&D Plan	Phase 1 (6 months): create a proof-of-concept based on Korean legal sources and the Korean language. This means that we will adapt our current proof-of-concept and formalization and check feasibility to use this in the Korean context. If a ground truth dataset is available, a small evaluation of the performance can be performed. Key tasks: - Define a use case together with the Korean SME. This includes both the Korean legislation they are targeting and their intended use or improvement for the SME. - Create a formalization for the Korean legal text, - Get a set of questions and ground truth answers (connected to the legal text), - Evaluate performance of the AI assistant on the Q&A set, - Write report on the findings. Milestones: - Formalization finished (3 months) - Evaluation on the Q&A set finished (5 months) - Report finished (after 6 months); if successful, a joint scientific publication can be written. Phase 2 (2y): create a well-performing AI Assistant for Korean. We evaluate and improve the proof-of-concept on Korean legal sources, using one or more Korean use cases. Key tasks: - Create an AI assistant front-end to be able to interact with (3 months) - Further improve performance (using the newest K-RAG methods, improve formalization based on findings) (1y): To improve performance: use different use cases: collect legal sources, formalized them, validate performance on ground truth data - Set-up a user experiment (in year 2): collect legal sources, gather participants, perform experiments (incl. logging data and questionnaires), evaluate performance and refine - write report on the findings (after each year) Milestones: - AI Assistant v1 finished (3 months) - AI Assistant v2 finished (1y) - Report with findings on use cases finished (1y) - AI Assistant v2 finished (2y) - Report with findings on user experiment finished (2y) Both reports could be converted to scientific papers, if applicable – the details that will be shared shall be discussed with the SME before starting a paper. Depending on the number of use cases and the result, the project could be extended to further improve performance. To quick-start the project, we use our knowledge on this topic (references 1-9) and our code base (10) as a head start.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Roles and Responsibilities of the Research Institute: - Technical coordination of the project - Development of key technology: programming and setting up the AI assistant - Formalization design and development, incl. knowledge sharing of how to use it. - Code Infrastructure: usage of GPUs and access to interface to use the assistant - Benchmarking and performance analysis of the AI assistant and formalization - Program management: project planning, milestone tracking, risk management, overall coordination of task. -
Expected R&R of Korean SME Partner	Roles and Responsibilities of the Korean SME: - Validation of the results in a specific use case - Exploitation of the results, including pathways to commercialization In Phase 1, the SME provides: - Use case definition: the use case / intended use of the technology - Regulatory source: a Korean legal source - Knowledge elicitation: questions and answers based on the legal sources - Legal validation: legal expertise on the legislation, which is used as validation - Performance benchmarking support: help in interpreting the results to further analyze the performance In Phase 2, the SME provides: - Use case expansion: more use cases / intended uses of the technology - Regulatory corpus extension: a set of Korean legal sources - Knowledge elicitation: question and answer on these legal sources - Experimental support: help in setting up the user experiment, mainly by providing the use case and collecting the participants - Legal Validation: legal expertise on the legislation, which is used as validation - help in interpreting the results to further analyze the performance - Technical feedback and system refinement: input to further develop the AI assistant, for example by providing feedback on the user interface to interact with - Exploitation strategy and commercialization: help in finding a partner to bring the AI Assistant to TRL level 9, which might include the (joint) commercialization of the technology in local markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes of the project are: - An improved formalization module for laws and regulations, that is suitable for at least Dutch and Korean - A formalized regulatory knowledge base: machine-readable representations of Dutch and Korean laws and regulations (which ones to be decided) - A TRL 7 AI-powered regulatory question answering module (the AI assistant), that can handle as least Dutch and Korean legal sources and both languages The impact of the project on the short term covers mainly effects on the organizational level, which are: - Reduced compliance time and costs and with that higher operational efficiency: lower manual interpretation efforts, lower expenses on legal expertise, reduced risk of

	misinterpretation and compliance errors, more quickly adapting to new laws and regulations, enabling more real-time compliance checking. The impact of the project on the longer term and its societal impact are: - Improving understanding and accessibility of legal knowledge and with that trust and transparency in the regulatory landscape. - Enabling innovation by easing compliance checking fosters the development of new products and services.

6. References

(1) Bakker, Roos, Maaike de Boer, Romy van Drie, and Daan Vos. 2022. "Extracting Structured Knowledge from Dutch Legal Texts: A Rule-Based Approach." *Companion Proceedings of the 23rd International Conference on Knowledge Engineering and Knowledge Management,* 3256 (September): 10. https://ceur-ws.org/Vol-3256/km4law1.pdf.

(2) Bakker, Roos, Romy van Drie, Maaike de Boer, Robert van Doesburg, and Tom Van Engers. 2022. "Semantic Role Labelling for Dutch Law Texts." *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, June, 448–57. https://aclanthology.org/2022.lrec-1.47.

(3) Bakker, Roos, Akke Schoevers, Romy van Drie, Marijn Schraagen, and Maaike de Boer. 2025. "Semantic Role Extraction in Law Texts: A Comparative Analysis of Language Models for Legal Information Extraction." *Artificial Intelligence and Law*, March 24, 1–35. https://doi.org/10.1007/s10506-025-09437-x.

(4) Drie, Romy van, Maaike de Boer, Roos M. Bakker, Ioannis Tolios, and Daan Vos. 2023. "The Dutch Law as a Semantic Role Labeling Dataset." *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law* (New York, NY, USA), ICAIL '23, September 7, 316–22. https://doi.org/10.1145/3594536.3595124.

(5) Redelaar, F., Van Drie, R., Verberne, S., & De Boer, M. (2024, November). Attributed Question Answering for Preconditions in the Dutch Law. In *Proceedings of the natural legal language processing workshop 2024* (pp. 154-165).

(6) van Drie, R., Bakker, R. M., Di Scala, D., de Boer, M. (2026), QuALA-NL: Question & Answer with Legal Attribution in Dutch, under review.

(7) Doesburg, Robert van, Tijs van der Storm, and Tom van Engers. 2016. "Calculemus: Towards a Formal Language for the Interpretation of Normative Systems." *Artificial Intelligence for Justice (AI4J)*, 65–69. https://www.ai.rug.nl/~verheij/AI4J/papers/AI4J_paper_15_doesburg.pdf.

(8) Breteler, Jeroen, Thom van Gessel, Giulia Biagioni, and Robert van Doesburg. 2023. "The FLINT Ontology: An Actor-Based Model of Legal Relations." In *Knowledge Graphs: Semantics, Machine Learning, and Languages*. IOS

Press. https://doi.org/10.3233/SSW230016.

(9) Doesburg, Robert van, and Tom van Engers. 2019. "Explicit Interpretation of the Dutch Aliens Act." In *Proceedings of the Workshop on Artificial Intelligence and the Administrative State*, edited by Luther Karl Branting, vol. 2471. CEUR Workshop Proceedings. CEUR-WS.org. http://ceur-ws.org/Vol-2471/paper6.pdf.

(10) Breteler, Jeroen, Thom van Gessel, Giulia Biagioni, et al. 2024. "Normative Systems · GitLab." GitLab, January 17. https://gitlab.com/normativesystems.

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Maaike de Boer (PhD) is a senior Scientist at TNO within the Data Science department. She has a background in AI and linguistics (Msc 2013, PhD 2017). At TNO Maaike mainly focuses on text mining / natural language processing (NLP) and hybrid AI: combining data-driven methods and knowledge-driven methods (such as ontologies) to get the best of both worlds. Projects include scientific projects and European or Horizon2020 projects such as TrustLLM and Cyclopes, but also Dutch and B2B projects such as the Growth Fund project Vaardig met Vaardigheid and Norm Engineering. Her projects are mainly focused in the defence, labour market and health domains. In previous years she has worked on NLP techniques for strategic foresight. Further details can be found through LinkedIn and Google Scholar.

Stephan Raaijmakers is a senior scientist at TNO, specialized in machine learning-based natural language processing (NLP), and a professor of Communicative AI at Leiden University, where he focuses on responsible use of Large Language Models in communicative user interfaces. He is the lead scientist of GRAIL, a TNO initiative for Responsible Generative AI. He has ample Horizon Europe experience, a.o. as the Technical Coordinator of two Horizon Europe projects. Stephan has authored two books on deep learning-based NLP and LLMs. In recent years, he has contributed to TNO projects on machine learning-based trend analysis of textual data.

(max 10 pages)

Freek Bomhof (MSc) is senior consultant at TNO within the Data Science department. He graduated in 1990 from the University of Twente on the topic of Digital Pattern Recognition. He then worked as a researcher, project manager and programme manager at KPN Research on sorting systems that read handwritten addresses on postal letters. From 1997 he was (deputy) R&D manager at the System Integration & Multimedia department. In 2003 he joined TNO as senior project manager, mainly on projects where complex business processes are supported with innovative ICT; notable projects include the introduction of self-steering for engineering teams within KPN including an innovative information system to support this social innovation.

Freek became more interested in the way individuals and organisations adopt ICT innovations, and what role these can play for aspects like sustainability, or how these can help SMEs to stay competitive. He has developed and led numerous projects on the topic of ICT and Sustainability, for instance he was involved as research coordinator in the first phase of the Amsterdam Smart City programme. Freek has worked for many SMEs and SME associations, mainly the engineering sector and building & construction sector. He has led a large European research project on the application of BIM in healthcare design.
Since 2013 Freek works in the Data Science department at TNO, concentrating on applications of data science and Artificial Intelligence in the Defence, Safety & Security domain (and also in other domains), both in early research, collaborative international R&D (EU H2020, ITEA) and close-to-market contract research. With a background in data science, Freek can identify opportunities for applications of research results, while at the same time leveraging his long experience in other fields to make connections to other relevant domains. Multidisciplinarity is the key to the success of any innovation, and this is maybe even more true for data-driven innovations.

Freek is an evaluator for Horizon 2020 project proposals, teaches IPMA internally (D level), is active as taskforce lead in the Big Data Value Association and is a regular and enthusiastic speaker at national and international conferences.

Roos Bakker (MSc) Roos Bakker (MSc) is a data scientist at TNO within the Data Science department, and a PhD student at Leiden University. She obtained an MSc in Artificial Intelligence from Utrecht University in 2018 and a BA in linguistics in 2015. In 2022, she started a PhD in collaboration with TNO and Leiden University on the topic of Knowledge Graph Extraction, Enrichment, and Evaluation. Roos is experienced in manual ontology development and Natural Language Processing, and is passionate about combining both to advance the field of knowledge representation and extraction. She is also ardent about developing code in an open and structured manner, examples can be found in open-source repositories such as Normative systems · GitLab, Covid-19 Ontology on Test Information, Common Greenhouse Ontology . Further details can be found at https://www.linkedin.com/in/roos-bakker and Roos Bakker - Google Scholar

Saskia Lensink is a senior consultant and business developer and specializes in language and speech technologies. She applies her knowledge of NLP and ASR in various projects, and is active in a diverse set of consortia and networks to promote sovereign and high-performing European large language models. She is currently the product owner of GPT-NL

Saskia has a background in linguistics. She earned her PhD in Leiden University. As a linguist, she examines the structures within language. She measures how the human brain, ears, and speech apparatus work. She also researches how this can be captured in computer models (natural language processing and computational linguistics). As a product owner at GPT-NL, Saskia brings the various teams and the external world together. This includes data acquisition, system architecture, and data curation, as well as the requirements and interests of external stakeholders.

Robert van Doesburg is a senior scientist at TNO/Leibniz Institute and a PhD candidate at the University of Amsterdam, where he conducts research on normative systems for humans and machines. His work focuses on developing methods and tools that make norms explicit, interpretable, and operational for large organisations. At TNO, he leads the norm-engineering programme, which employs the Calculemus protocol and the Flint language to model and automate normative reasoning. This programme is supported by several Dutch ministries and aims to enable large-scale, legally robust applications of AI for interpreting regulatory texts.

(max 10 pages)

Before joining TNO, Van Doesburg worked for more than fifteen years at the Dutch Immigration and Naturalisation Service (IND), holding various project and programme management roles, including contributions to the rule-based information system INDIGO and organisational restructuring efforts. Earlier in his career, he served as a consultant at Andersson Elffers Felix, carrying out projects for ministries such as the Ministry of the Interior and Kingdom Relations, the Ministry of Justice and Security, and the Council of State. His professional trajectory reflects a sustained focus on normative analysis, policy execution, and the development of systems capable of handling legal and administrative complexity at scale.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization